

MR_IL6R_CRC v45 Independent GEO Validation Rescue Report

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Purpose: verify the uploaded v44 GEO validation outputs, explain why the validation results cannot yet be integrated, and provide a corrected v45 rerun workflow.

Executive conclusion

The uploaded v44 independent validation run does not yet contain usable GEO validation result tables or figures. The GEO download step completed, but the extraction step failed when `Biobase::exprs()` was applied to a character object rather than an `ExpressionSet`. Therefore, it would be scientifically unsafe to write GEO validation results into the Results/Discussion at this stage.

Evidence from uploaded files

Item	Observed status	Interpretation	Action
PowerShell log	Download completed; extraction failed with <code>exprs(character)</code> .	No validation tables can be trusted yet.	Run v45 hotfix.
v44_independent_validation.zip	Contains four tiny RDS files and empty results folder.	No biological validation result available.	Regenerate with robust scripts.
Manuscript integration	Requested v45 integration cannot be completed honestly yet.	Avoid unsupported claims.	Use insertion template after rerun.

v45 rescue workflow

- 01_download_GEO_validation_cohorts_v45.R: re-downloads GEO cohorts and writes a download audit.
- 02_extract_IL6R_and_validate_GEO_cohorts_v45.R: handles `ExpressionSet`, list-of-`ExpressionSet` and character-path objects robustly.
- 03_make_GEO_validation_supplementary_figure_v45.R: creates a supplementary validation overview figure after result tables are generated.
- Manuscript integration template: provides conservative wording after successful rerun.

Run commands

```
# (PowerShell) 运行 v45 independent GEO validation hotfix
$ROOT = "D:\SCI2\MD-1\MR_IL6R_CRC_project_template_v4"
$PKG = Join-Path $ROOT "2_code\99_reporting\v45_independent_validation_rescue"
cd $PKG
```

```
powershell -ExecutionPolicy Bypass -File ".  
\\01_R_hotfix_scripts\run_v45_independent_validation_hotfix.ps1" -ROOT $ROOT  
echo "__DONE__ $(Get-Date)"
```

Decision for JTM revised package

This v45 package is not a final revised JTM package. It is a validation-rescue and integration-ready package. Once the corrected v45 output is uploaded, the next step should be v46: integrate actual GEO validation values into Results, Discussion, Supplementary Tables and Supplementary Figure, and rebuild the JTM revised submission package.